Chan Hur

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LinkedIn | GitHub Blog | Google Scholar

Research Interests

Multimodal learning and representation learning, with a focus on vision-language models, text–video retrieval, GUI agents, compositional image retrieval, video question answering, and applications of large language and multimodal models. Research interests include cross-modal alignment, generative caption utilization, bridging semantic gaps between modalities, and domain-specific AI for medical applications.

Research Experience

Researcher, ETRI 2026.02–Present

- Digital Convergence Research Department
  Electronics and Telecommunications Research Institute
  Research on vision-language models, multimodal retrieval, and GUI agents.

Education

Ph.D. | *Department of Computer Science and Engineering* 2021.03–2026.02
- Kyungpook National University Daegu, South Korea
- Dissertation: Advancing Text-Video Retrieval with Frame-Wise Generative Information
- Advisor: Prof. Hyeyoung Park

M.S. | *Department of Computer Science and Engineering* 2019.03–2021.02
- Kyungpook National University Daegu, South Korea
- Thesis: Zero-Shot Image Recognition Model Using Image Description
- Advisor: Prof. Hyeyoung Park

B.S. | *Department of Mechanical Engineering* 2014.03–2019.02
- Kyungpook National University Daegu, South Korea

Publications

International Conferences

- **Surface-Attribute Text Conditioning for RGB-to-HSI Reconstruction**
  Chan Hur, Woonggyu Min, Ho-Youl Jung, Soo Jun Park, Myungeun Lim
  Asian Conference on Computer Vision (ACCV), 2026. Accepted.
- **Prescription Orders as a Prediction-Time Signal for Forward Acute Kidney Injury**
  Hyung Wook Noh, **Chan Hur**, Jeonghyun Kim
  IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI), 2026. Accepted.
  https://openreview.net/forum?id=xfrk4jbxVa
- **PopupBench: Diagnosing Popup Blindness in Vision–Language GUI Agents**
  Chan Hur*, Youngjae Lee*, Hyeyoung Park, Hyung Wook Noh, Youngwan Lee
  NeurIPS 2026 Workshop on Interpreting Agent Behavior (IAB). Accepted.
- **Semantic Hardness Is Not Visual Hardness: Sign-Aware Hard Negative Mining for Sign Language Retrieval**
  Junmyeong Lee, **Chan Hur**, ChangSu Choi, Sukmin Cho, Fitsum Gaim, Eui Jun Hwang, Hoyun Song, KyungTae Lim
  Annual Meeting of the Association for Computational Linguistics (ACL), 2026.
- **ReFineVQA: Iterative Refinement of Video Description via Feedback Generation for Video Question Answering**
  Jeongwan Shin*, **Chan Hur***, Seongmin Cho, Jae-Ho Choi, Hyeyoung Park
  Winter Conference on Applications of Computer Vision (WACV), 2026.
- **Narrating the Video: Boosting Text-Video Retrieval via Comprehensive Utilization of Frame-Level Captions**
  Chan Hur*, Jeong-hun Hong*, Dong-hun Lee, Dabin Kang, Semin Myeong, Sang-hyo Park, Hyeyoung Park
  Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
  https://arxiv.org/abs/2503.05186
- **Local Feature Aggregation with Attribute Anchors for Text-guided Image-Text Retrieval**
  Chan Hur, Hyeyoung Park
  International Conference on Artificial Intelligence in Information and Communication (ICAIIC), 2025.
  https://ieeexplore.ieee.org/document/10920824
- **Automatic Image Recommendation for Economic Topics using Visual and Semantic Information**
  Chan Hur, Changhun Hyun, Hyeyoung Park
  International Conference on Semantic Computing (ICSC), 2020.
  https://ieeexplore.ieee.org/document/9031480

International Journals

- **EENet: Embedding Enhancement Network for Compositional Image-Text Retrieval using Generated Text**
  Chan Hur, Hyeyoung Park
  Multimedia Tools and Applications, 2023. SCIE (Q1).
  https://doi.org/10.1007/s11042-023-17531-y
- **Zero-shot Image Classification using Captioning-based Rectified Embedding Vector**
  Chan Hur, Hyeyoung Park
  Applied Sciences, 2023. SCIE (Q2).
  https://doi.org/10.3390/app13127071
- **An Image Selection Framework for Automatic Report Generation**
  Changhun Hyun, **Chan Hur**, Hyeyoung Park
  Multimedia Tools and Applications, 2022. SCIE (Q1).
  https://doi.org/10.1007/s11042-022-13120-7

Preprints & Under Review

- **Forward Clinical Features Enable Treatment-Aware Prediction and What-If Scenario Analysis for Acute Kidney Injury**
  Hyung Wook Noh*, **Chan Hur***, Jeonghyun Kim*, Yoonseon Song, Hyunsoo Cho, Yunseo Ku, Dae Eun Choi
  Under review, 2026.
- **One Prior, Many Cameras: Retrain-Free Spectral Reconstruction with Calibrated Metamer Uncertainty**
  Woonggyu Min, **Chan Hur**, Ho-Youl Jung, Soo Jun Park, Myungeun Lim
  Under review, 2026.
- **PopupBench: Revealing Popup Blindness in Vision-Language GUI Agents**
  Chan Hur, Youngjae Lee, Hyeyoung Park, Youngwan Lee
  Under review, October 2026.
- **TraceCIR: Executable Intent Trajectories for Stateful Language Grounding in Multi-Turn Image Retrieval**
  Chan Hur, Minkyung Cho, Jun Seo Kim, Won Jun Oh, Hyung Wook Noh, KyungTae Lim
  Under review, October 2026.
- **MELON: A Large-Scale Dataset for Multi-Event Text-to-Long-Video Retrieval**
  Chan Hur*, SeungWoo Song*, Jeong-hun Hong*, Won Jun Oh, Hyeyoung Park, KyungTae Lim
  arXiv preprint arXiv:2609.01654, 2026.
  https://arxiv.org/abs/2609.01654

Domestic Publications

- **Improving text-video retrieval performance using video features combined with generation-based caption information**
  Chan Hur, Dong-hun Lee, Sang-hyo Park, Hyeyoung Park
  KIISE Transactions on Computing Practices, KCI, 2024.
- **Feature Vector Combination on Joint Embedding Space using Captions for Text Video Retrieval**
  Chan Hur, Dong-hun Lee, Sang-hyo Park, Hyeyoung Park
  Korea Computer Congress (KCC), 2023. **Best Paper Award.**
- **Deep learning-based Lightweight Model for Carbon-Reduction Activity Image Classification**
  Hojun Song, Yunhee Koo, Sehyun Park, Jeoungwon Cha, Youngjae Lee, **Chan Hur**, Hyeyoung Park
  Journal of Embedded Systems and Applications, KCI, 2022.
- **A Study on the Alternative Method of Video Characteristics Using Captioning in Text-Video Retrieval Model**
  Dong-hun Lee, **Chan Hur**, Hyeyoung Park, Sang-hyo Park
  Journal of Embedded Systems and Applications, KCI, 2022.
- **A study on Video-text Retrieval model using Caption Information**
  Dong-hun Lee, **Chan Hur**, Hyeyoung Park, Sang-hyo Park
  Korea Artificial Intelligence Conference, 2022.

Patents

전자 장치 및 그의 이미지를 추천하는 방법 2025.05

- Registered (등록).

언어 생성모델을 이용한 대화형 이미지 추천 시스템 2023.06

- Patent application (출원).

텍스트 기반 비디오 검색 장치 및 그 비디오 검색 방법 2025.05

- Registered (등록).

Awards

Best Graduate Ph.D. Student Award 2026

- Department of Computer Science and Engineering, Kyungpook National University

Best Paper Award 2024

- LSTM 기반 온실 환경 예측 및 제어 시스템 최적화
  ICT 대학생 논문 경진대회

Best Paper Award 2023

- Feature Vector Combination on Joint Embedding Space using Captions for Text Video Retrieval
  Korea Computer Congress (KCC)

Lectures

Deep Learning and Machine Learning with Python 2023–2025

- Kyungpook National University IT Education Center

Introduction of Generative AI and Applications 2024–2025

- Kyungpook National University IT Education Center

Machine Learning 2023–2025

- Korea National Open University

Skills

Deep Learning Frameworks: PyTorch, TensorFlow
Languages: Korean (native), English (fluent), Japanese (intermediate)